Expert Systems With Applications 290 (2025) 128429



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa





Dynamic selection of ensemble-based regression models: Systematic literature review ⋆

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ARTICLE INFO

Keywords:
Machine learning
Ensemble
Dynamic ensemble
Dynamic selection

ABSTRACT

Dynamic selection techniques have emerged as a promising approach to enhance ensemble-based regression methods, offering adaptive solutions for dynamic and heterogeneous data. This study presents a comprehensive Systematic Literature Review (SLR) on dynamic selection in regression models, aiming to investigate how these techniques can improve predictive effectiveness across various contexts. The review encompassed 21 primary empirical studies published between 2015 and 2024, addressing six research questions (RQs) covering models used, evaluation metrics, implementation strategies, and performance criteria. The findings highlight the use of 59 distinct models and 26 evaluation metrics, such as RMSE, MAE, and R^2, applied across 144 datasets from domains including healthcare, energy, finance, and time series forecasting. Four main categories of dynamic selection techniques were identified: Dynamic Selection (DS), Dynamic Weighting (DW), Dynamic Weighting with Selection (DWS), and Optimization-Based Methods. These approaches demonstrated significant improvements in accuracy, robustness, and scalability, particularly in scenarios involving noise, data imbalance, and concept drift. However, challenges such as high computational costs and the lack of unified evaluation frameworks remain. This study provides a consolidated foundation for researchers and practitioners seeking to explore and apply dynamic selection techniques, highlighting promising avenues for future research, including integration with deep learning and distributed systems.

1. Introduction

In recent years, ensemble learning has become one of the most prominent techniques in the field of Machine Learning (ML), offering improved accuracy, robustness, and generalization by combining multiple models (Dietterich, 2000; Ganaie et al., 2022; Zhou, 2012). These models can be grouped in homogeneous ensembles (using the same learning algorithm) or heterogeneous ensembles (combining different types of algorithms), and are applied in both classification and regression tasks.

Ensemble construction generally involves three key stages: generation, selection, and combination. In the generation phase, diversity is introduced among base models via methods such as Bagging (Breiman, 1996) and Boosting (Tu et al., 2017). The selection stage aims to choose the most relevant models, often using cross-validation or performance-based criteria. In the final stage, predictions from selected models are combined using strategies like averaging or weighted voting to enhance performance (Allende & Valle, 2017).

While static ensembles, such as those produced by Bagging or Boosting, offer substantial improvements over single-model approaches, they assume that the same set of base models and combination strategy will remain optimal across the entire input space (Breiman, 1996; Dietterich, 2000). This assumption may not hold in real-world scenarios where the data distribution is heterogeneous, dynamic, or evolves over time. In such contexts, static ensembles can become suboptimal, as they are unable to adapt to local variations or concept drift. This limitation motivates the adoption of Dynamic Ensemble Selection (DES) techniques, which tailor the model selection or weighting process to each individual instance based on localized performance estimations.

Dynamic Ensemble Selection extends this approach by selecting or weighting models at prediction time based on their competence in the local region of the feature space (Cruz et al., 2015). This dynamic behavior allows ensembles to adapt to input variability, which is especially beneficial in nonstationary environments or tasks with high data heterogeneity.

⋆ This document is the results of the research project funded by the National Science Foundation.

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https://doi.org/10.1016/j.eswa.2025.128429
Received 27 March 2025; Received in revised form 18 May 2025; Accepted 30 May 2025
Available online 1 June 2025
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Strategies such as **Dynamic Selection (DS)**, **Dynamic Weighting (DW)**, and **Dynamic Weighted Selection (DWS)** have been widely employed in the SDE literature, both for classification and regression (Britto et al., 2014; Cruz et al., 2018; Oliveira et al., 2022; Sun et al., 2022). These approaches explore different ways of adjusting the ensemble to the local context: selecting the most competent model (DS), weighting all models proportionally to their competence (DW), or combining both ideas (DWS). Finally, **Optimization-Based Methods** apply metaheuristics or mathematical formulations to optimize the selection or weighting process. This taxonomy provides a structured perspective on how dynamic selection has been implemented in regression tasks across different domains.

Several Systematic Literature Reviews (SLRs) have examined ensemble-based methods in domains such as software effort estimation (Ali & Gravino, 2019), defect prediction (Khanna, 2022), and general model selection (Cabral et al., 2023). While these studies highlight the effectiveness of ensembles, most focus on static selection or classification scenarios. A detailed and structured review focused specifically on dynamic selection techniques in regression is still lacking.

However, a structured and focused review of dynamic selection techniques specifically for regression tasks is still lacking in the literature. Classification problems have received considerable attention in this context, largely due to the availability of well-established competence measures and clearly defined decision boundaries (Britto et al., 2014; Cruz et al., 2019, 2015). In contrast, regression tasks introduce unique challenges, such as continuous outputs, variability in error distributions, sensitivity to local fluctuations, and the absence of a consensus on local competence estimation metrics (Moura et al., 2019; Wang et al., 2022). These factors make regression a more complex and less explored scenario for dynamic selection, justifying the need for dedicated investigation.

Recent studies have also explored practical applications of dynamic model selection across different domains, reinforcing the importance of this approach. For instance, dynamic combination strategies have been proposed for price prediction using multiple models and adaptive sliding windows (Hao et al., 2023); heterogeneous ensemble methods with overfitting-aware weighting for credit risk assessment (Xia et al., 2020); dynamic selection approaches adapted to nonstationary environments with concept drift detection (Wei et al., 2024); and empirical investigations on the impact of selection criteria such as diversity and similarity in dynamic ensemble systems (Lustosa Filho et al., 2018). These advances complement the existing literature by highlighting the applicability and flexibility of dynamic selection in diverse scenarios.

Within ensemble learning, Dynamic Selection techniques aim to tailor model selection to specific regions of the feature space. Each model specializes in a local area–called the Region of Competence (RoC)–often determined by K-Nearest Neighbors (kNN) Moura et al. (2019). Several studies have proposed refinements to these techniques to improve accuracy, such as selection based on local oracles or statistical discrimination measures (Pereira et al., 2018; Sani et al., 2022).

Despite the recent progress, current approaches in regression still lack standardized strategies, reproducible frameworks, and comprehensive taxonomies for dynamic model selection. Moreover, the scarcity of studies comparing dynamic selection strategies across domains and datasets further emphasizes the gap in the field.

In this context, the present study conducts a Systematic Literature Review (SLR) on dynamic selection techniques applied to ensemble-based regression. Our goal is to analyze the strategies used, identify common models and evaluation metrics, and map the application domains and performance assessment practices.

The main contributions of this paper are: (i) a structured synthesis of DES methods applied to regression; (ii) a classification of dynamic strategies into four types–Dynamic Selection, Dynamic Weighting, Dynamic Weighted Selection, and Optimization-Based Methods; and (iii) a discussion on practical applications and future directions including deep learning integration and concept drift handling.

The remainder of this paper is structured as follows: Section 2 presents the theoretical background; Section 3 details the methodology of the SLR; Section 4 discusses the findings; Section 5 addresses threats to validity; and Section 6 outlines the conclusions and directions for future research.

2. Background

The objective of this section is to present concepts that support this research, as well as related works.

2.1. Ensemble learning

Ensemble Learning (EL) researchers seek to answer two fundamental questions: (1) How to generate the ensemble? and (2) How to integrate the predictions of the ensemble models? The answers to these questions are essential to ensure that models complement each other and maximize predictive accuracy.

Formally, ensemble learning encompasses methods that generate a variety of models, which are then intelligently combined to provide a single prediction, applicable to both classification and regression problems (Mendes-Moreira et al., 2012). This approach has garnered considerable attention in recent years, with promising results being documented (e.g., Carvalho et al., 2020; Cabral et al., 2023; Minku & Yao, 2013).

Building an ensemble typically involves three main steps, as illustrated in Fig. 1.

1. **Generation:** The first step is to assemble a diverse set of candidate models, each offering a unique perspective on the problem. These models, denoted as $\hat{f}_1, \hat{f}_2, \ldots, \hat{f}_k$, are trained independently using different algorithms, data subsets, or feature representations to enhance diversity. The set of all generated models is represented as F_0.
2. **Selection:** Not all generated models contribute equally to ensemble performance. In this step, a subset of models $F \subseteq F_0$ is chosen based on specific criteria such as accuracy, diversity, or competence in different regions of the feature space. The selected models $(\hat{f}_1, \hat{f}_4, \ldots, \hat{f}_k)$ are called **ensemble components**, as they work together to improve overall predictive performance.
3. **Aggregation:** The final step consists of **combining** the predictions of the selected models into a single output. The function $\hat{f}_F(\hat{f}_1, \hat{f}_4, \ldots, \hat{f}_k)$ represents the aggregation process, which can be performed using techniques such as *majority voting*, *weighted averaging*, or *meta-learning*, depending on the application and the nature of the models involved. The objective of this step is to leverage the complementary strengths of the selected models to produce a more robust and accurate final prediction.

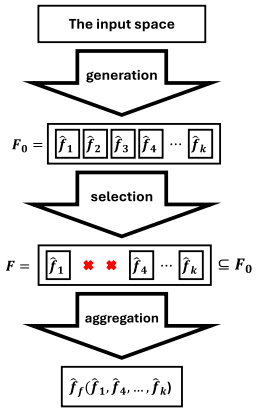


Fig. 1. Combined models approach. Adapted from Mendes-Moreira et al. (2012).

This structured approach ensures that the ensemble maximizes accuracy while minimizing redundancy among models, ultimately leading to improved generalization on unseen data.

A wide range of strategies can be employed to generate and select ensemble models. The effectiveness of an ensemble largely depends on the diversity and quality of its individual components, as well as the method used to integrate their predictions. Several strategies have been developed to generate and select ensemble models. Bagging (Bootstrap Aggregating) Breiman (1996) focuses on reducing variance by training models on different subsets of the data. Boosting sequentially trains models, assigning higher weights to previously misclassified instances to improve accuracy (Freund & Schapire, 1997). Stacked Generalization (Stacking) Carvalho et al. (2020) combines the outputs of multiple base models using a meta-learner to optimize prediction. Negative Correlation Learning (NCL) promotes diversity by encouraging base models to make uncorrelated errors (Minku & Yao, 2013). These techniques aim to enhance diversity and predictive performance across ensemble components.

Recent proposals have explored the combination of diverse ensemble components using dynamic or heterogeneous strategies. Xia et al. (2020), for example, proposed an ensemble composed of five tree-based models–Random Forest, Gradient Boosting Decision Trees (GBDT), XGBoost, LightGBM, and CatBoost–combined via an overfitting-cautious dynamic weighting scheme. Similarly, Hao et al. (2023) investigated the use of multiple base regressors in a dynamic ensemble approach, integrating them through adaptive combination strategies and a sliding-window mechanism designed for time-series forecasting. These contributions highlight the potential of dynamic and heterogeneous ensembles to improve prediction performance while maintaining robustness in complex domains.

In conclusion, ensemble learning has become a powerful tool for enhancing model performance by leveraging the strengths of diverse learners. Despite its advantages, the challenge remains: how can we efficiently select the best models for a specific task within the ensemble? Addressing this question is the core idea behind Dynamic Ensemble Selection. In the next chapter, we explore DES techniques, which dynamically select the most effective models for each instance, further enhancing the power and adaptability of ensemble learning.

2.2. Dynamic ensemble selection

Dynamic Ensemble Selection represents a shift from traditional, static ensembles by dynamically selecting only the most relevant models at prediction time (Cruz et al., 2020). This approach allows DES to better handle data variability, reducing the negative impact of models that may perform poorly on specific instances. As a result, DES improves the ensemble's adaptability and accuracy for diverse prediction tasks (Oliveira et al., 2022; Sani et al., 2022; Sun et al., 2022).

The DES process relies on selection criteria grounded in model competency, such as Local Accuracy, where models are evaluated within specific regions of the data, and Distance to Error, which assesses models based on proximity to correctly classified instances (Moura et al., 2019, 2021b; Pereira et al., 2018). By identifying models that perform well in particular regions of the input space, DES ensures that only the most competent models are chosen for each prediction. Additionally, techniques such as weighted distances (Xu et al., 2024) and Fuzzy Inference Systems (FIS) Kurzynski et al. (2018) have been proposed to enhance the DES framework by incorporating nuanced measures of model reliability.

Complementing these efforts, Lustosa Filho et al. (2018) conducted an empirical analysis on the impact of different selection criteria–particularly diversity and similarity–on the performance of DES methods. Their results suggest that alternative distance-based measures can significantly influence the effectiveness of model selection, offering guidance for improving ensemble performance in varying contexts.

Unlike static ensembles, which include all models regardless of their suitability for each prediction, DES adapts dynamically, which has shown to increase predictive performance, particularly in complex and variable datasets (Cabral & Oliveira, 2021; Moura et al., 2021a). This adaptability is especially beneficial in fields with significant data shifts or recurring patterns, as seen in efforts to predict short-term load (Yang et al., 2023) and software development effort estimation (Cabral et al., 2017).

In contexts characterized by data drift and nonstationary distributions, adaptive DES frameworks have also been explored. Wei et al. (2024) introduced an approach called AB-DES (Adaptive Bagging-based Dynamic Ensemble Selection), which integrates concept drift detection, resampling strategies, and local model selection using sliding windows. Although applied to classification tasks, this approach demonstrates the viability of combining adaptive mechanisms with dynamic selection in environments where data distributions evolve over time.

DES has proven advantageous in fields requiring high adaptability, such as time-series forecasting and anomaly detection (Liu & Zio, 2016; Oliveira et al., 2022). However, it introduces challenges, including increased computational costs due to runtime model selection and the need for robust data structures to manage local model competencies across data regions (Wang et al., 2022). These complexities must be carefully balanced to fully leverage the benefits of DES in real-time applications.

In conclusion, while DES has shown promising results in enhancing model adaptability and precision across various fields, it remains an area ripe for further exploration, especially in dynamic contexts and under changing data distributions. The following section delves into related works that have laid foundational insights into DES and Ensemble Techniques (ET), including advancements and remaining research gaps.

2.3. Related works

The studies listed in Table 1 encompass a range of approaches related to ensemble learning, dynamic selection, and regression tasks. The first three studies (Cabral et al., 2023; Idri et al., 2016; Khanna, 2022) focus on ensemble-based software development effort estimation and defect prediction, providing insights into how ensemble methods improve predictive performance in software engineering applications.

Despite the extensive use of ensembles in software engineering, their application in regression tasks remains relatively less explored. Moura et al. (2019) provide a foundational study on competence measures for dynamic regressor selection, investigating how different metrics can guide the model selection process.

Building upon this foundation, Oliveira et al. (2022) apply dynamic selection techniques to time series forecasting, demonstrating how adaptive model selection can enhance prediction accuracy in non-stationary environments. Similarly, Wang et al. (2022) propose a dynamic outlier ensemble to improve industrial process predictions, tackling challenges such as data variability and concept drift.

Table 1
Summary of related works on dynamic ensemble selection and regression.

Work	Goal
Idri et al. (2016)	Systematic review of software development effort estimation techniques based on ensembles.
Cabral et al. (2023)	Updated the 2016 SLR Idri et al. (2016) with new findings on ensemble effort estimation between 2016 and 2020.
Khanna (2022)	Systematic review on ensemble techniques for software defect and change prediction.
Moura et al. (2019)	Evaluating competence measures for dynamic regressor selection.
Oliveira et al. (2022)	Proposed a hybrid system based on dynamic selection for time series forecasting.
Wang et al. (2022)	Used a dynamic outlier ensemble to predict molten steel temperature.
Our Work	Systematic literature review on Dynamic Selection applied to regression tasks.

Next, we provide a detailed discussion of each of these works, highlighting their contributions and relevance to this study.

Idri et al. (2016) conducted a SLR aimed at analyzing software development effort estimation (SDEE) techniques based on ensemble effort estimation (EEE) from six perspectives: individual models used to build ensembles, accuracy of ensemble estimates, rules used to combine individual estimates, accuracy comparison between EEE techniques and individual models, accuracy comparison between different EEE techniques, and methodologies employed in constructing ensembles. The SLR was performed on studies published between 2000 and 2016, with an automated search in four databases (IEEE Xplore, ACM Digital Library, ScienceDirect, and Google Scholar). Initially, 4905 articles were found, plus five additional articles known to the authors, totaling 4910 articles. After applying the inclusion and exclusion criteria, 24 articles were selected for detailed analysis.

The study contributed in several ways, such as identifying the most commonly used ensemble techniques in SDEE, evaluating the overall accuracy of EEE techniques, demonstrating that they are generally more accurate than individual models, and identifying the methodologies used to select the base models in constructing ensembles, among others. Finally, the authors concluded that EEE techniques can be categorized as homogeneous and heterogeneous, with machine learning techniques being the most frequently used in constructing ensembles. The review revealed that EEE techniques generally offer acceptable estimation accuracy and are more accurate than individual models. The study emphasizes the usefulness of ensembles in overcoming the weaknesses of individual models, providing more precise and stable effort estimates.

The work of Cabral et al. (2023) updated the 2016 systematic review (Idri et al., 2016) with new findings from studies published between 2016 and 2020. The review aimed to collect and rigorously evaluate the literature on software effort estimation techniques (EEE) based on ensemble methods, addressing the limitations of individual methods. Initially, 3682 articles were retrieved using automatic search techniques in three databases (Scopus, IEEE Xplore, ACM Digital Library). After applying inclusion and exclusion criteria, 30 articles were selected for data extraction and analysis.

The updated review provided new evidence on using EEE techniques in software development, reinforcing previous findings and highlighting research opportunities in Dynamic Ensemble Selection, which remains underexplored. The study emphasized that, although ensemble techniques offer acceptable and superior estimation accuracy compared to individual models, no EEE technique stands out over the others. Machine learning techniques remain prevalent in the construction of ensembles. Among other findings, the review study encouraged the use of dynamic model selection as a promising area for future research.

In 2022, Khanna (2022) conducted a systematic literature review to evaluate the use of ensemble techniques for Software Defect Prediction (SDP) and software change prediction (SCP). The primary objective of the review was to systematically collect and rigorously evaluate literature that develops classification models using ET to predict defect-prone and change-prone parts of software. The review covered studies published from January 2000 to December 2020. The initial search identified 182 relevant studies, and after applying inclusion and exclusion criteria, 77 primary studies were selected for detailed analysis based on category, application, rules of formulation, performance, and possible threats of the proposed/utilized ensemble techniques.

The results indicated that ensemble techniques can significantly enhance the predictive performance for SDP and SCP. They were also found effective in several applications, such as their use as a learning algorithm for developing SDP/SCP models and addressing the class imbalance issue. The study emphasizes the need for further empirical research to explore under-researched Ensemble Techniques (ET), suggests improving ET methodologies, and highlights the importance of developing tools to support the practical implementation of ET in software engineering.

Moura et al. (2019) conducted a study evaluating competence measures for dynamic regressor selection. The objective was to analyze which criteria are most effective in determining which models should be dynamically activated during the inference phase. The study compared different competence metrics, exploring how model selection based on error, variability, and other factors can improve the accuracy of dynamic ensembles. The results highlighted the need for more robust competence estimation mechanisms, particularly in regression settings where traditional classification-based measures may not be directly applicable.

Oliveira et al. (2022) proposed a hybrid system based on dynamic selection for time series forecasting. The study explored how different models can be dynamically selected to predict patterns in time series data, improving forecasting accuracy. The approach combined traditional ensemble learning methods with dynamic strategies, ensuring greater adaptability to changes in time series patterns. Experiments showed that the proposed technique outperformed static approaches in complex forecasting scenarios, particularly in cases involving non-stationary data.

Wang et al. (2022) introduced a dynamic outlier ensemble for predicting molten steel temperature in industrial processes. The study focused on identifying outliers within the regression model set and dynamically adapting models to improve prediction accuracy. The proposed approach used dynamic model selection techniques to handle fluctuations in data quality, ensuring that only the most reliable models were used at each moment. Their findings demonstrated that adaptively selecting models based on data quality variations can significantly improve performance in industrial forecasting tasks.

3. Research methodology

This study adopts a systematic review process to examine the existing research on Dynamic Ensemble Selection applied to regression tasks. Following the guidelines proposed by Kitchenham and Charters (2007), the review was structured into a set of rigorous and predefined steps to ensure comprehensive coverage of relevant studies and objective synthesis. According to these guidelines, a systematic approach must be guided by a formal protocol with well-defined procedures to identify, evaluate, and interpret the available evidence from Primary Studies (PS) related to specific research questions. The main goal of a Systematic Literature Review (SLR) is to consolidate results from individual studies, assess their consistency, and identify knowledge gaps.

The review protocol developed for this study includes eight sequential steps: (i) definition of the **Research Questions**, (ii) development of the **Search Strategy**, (iii) selection of relevant **Literature Resources**, (iv) **Study Identification**, (v) application of **Inclusion and Exclusion Criteria**, (vi) **Quality Assessment**, (vii) structured **Data Extraction**, and (viii) comprehensive **Data Synthesis**.

An overview of these steps is formally summarized in Algorithm 1, while a visual representation is presented in Fig. 2.

Algorithm 1 Systematic literature review protocol.

1: Define research questions
2: Develop search strategy
3: Select literature resources
4: Identify studies from digital libraries
5: Apply inclusion and exclusion criteria
6: Assess study quality using a checklist
7: Extract data using structured extraction forms
8: Synthesize results and report findings

In the first step, a set of research questions was raised and defined based on the objective of this SLR. Considering these research questions, a search strategy was developed in the second step to identify relevant

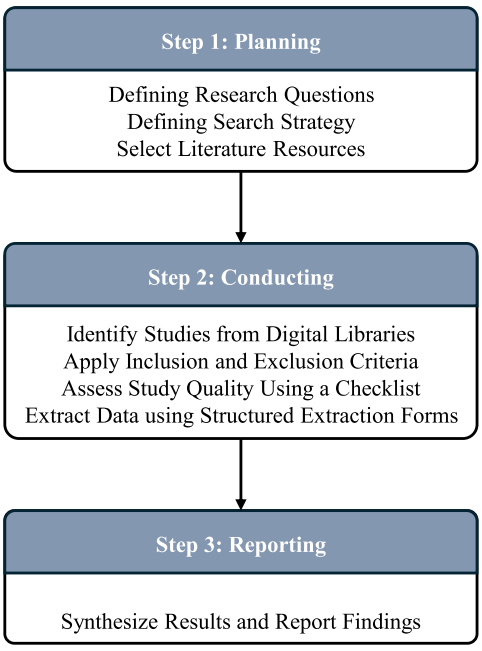


Fig. 2. Structured flowchart of the systematic literature review protocol.

studies that contribute to answering the research questions. This step involves determining the search string and selecting the bibliographic resources necessary for an iterative and impartial search process. In the third step, inclusion and exclusion criteria were established to determine which studies to include and which to discard. This step involves identifying all relevant studies from the selected literature resources based on the research questions.

Next, the relevant studies underwent a quality assessment process, during which a series of quality checklists was developed to facilitate evaluation. The last two steps involve data extraction and synthesis, respectively. In the fifth step, data extraction, all essential information related to the research questions was collected and stored in a table (referred to as the data extraction form). Finally, in the data synthesis step, the extracted data were analyzed and synthesized based on the research questions.

In any SLR, developing a review protocol is crucial to minimizing the risk of research bias and ensuring the rigor and repeatability of the study. The following subsections describe in detail all the steps of the SLR, and at the end of this section, an analysis of the threats to the validity of the review protocol is presented.

3.1. Systematic review protocol steps

Following the execution of the search strategy, a systematic and structured sequence of steps was performed to ensure the rigor and replicability of this review process. These steps, aligned with Kitchenham and Charters' guidelines (Kitchenham & Charters, 2007), are detailed below:

1. **Definition of Research Questions**: Established the main research objective and decomposed it into six guiding research questions (Section 3.2.1).
2. **Development of Search Strategy**: Constructed search strings combining key terms with logical operators and adapted them for each selected database (Section 3.2.2).
3. **Selection of Literature Resources**: Selected five relevant digital libraries: ACM Digital Library, IEEE Xplore, ScienceDirect, Scopus, and Web of Science (Section 3.2.4).
4. **Study Identification**: Conducted automated searches across all libraries using the defined search strings, targeting titles, abstracts, and keywords (Section 3.2.5).
5. **Application of Inclusion and Exclusion Criteria**: Screened all retrieved studies based on predefined inclusion and exclusion criteria to identify relevant primary studies (Section 3.3.1).
6. **Quality Assessment of Studies**: Evaluated the methodological rigor of selected studies using a standardized checklist comprising nine quality questions (Section 3.3.2).
7. **Data Extraction**: Extracted key information related to research questions from each selected study using a structured data extraction form (Section 3.3.3).
8. **Data Synthesis**: Aggregated and synthesized the extracted data, presenting the findings through visualizations and tables (Section 3.4.1).

With the overall protocol steps established, the following subsections describe each phase in greater detail.

3.2. Planning

The planning phase of the SLR involved several key steps: defining research questions and defining search strategy. This search strategy comprised three elements: constructing search strings, selecting appropriate literature resources, and establishing a well-defined search process.

3.2.1. Research questions

The aim of this work is to analyze, identify, clarify and evaluate the empirical evidence from various studies that use Dynamic Selection methods applied to regression. The analysis is guided by the research questions. To achieve this, the main Research Question (RQ) was defined: How can Dynamic Selection be used to improve the effectiveness of Regression methods? In particular, which dynamic selection criteria for regression models result in better prediction accuracy? To address this research question, it was further divided into six sub-questions:

RQ1: *Which individual models were used in the dynamic selection, composition, and combination?*
This question seeks to identify which specific regression models were used as individual components within the dynamic selection framework. Dynamic model selection involves choosing or combining different models based on real-time data characteristics to improve predictions' accuracy and adaptability. Identifying the individual models used helps to understand the diversity and robustness of the system.

RQ2: *What metrics were used to measure the competency of the regression models in different regions of the data space?*
This question focuses on how metrics are used to evaluate model competency across different regions of the dataset, better capturing the concept of a "region of competency" within the data space.

RQ3: *What datasets were used and how were they applied in the dynamic selection of regression models?*
This question seeks to understand which specific datasets (areas of knowledge) were used in studies or experiments involving dynamic regression model selection.

RQ4: *What are the different dynamic selection techniques used in conjunction with regression methods?*
This question seeks to understand the strategies and methods used to implement dynamic model selection, particularly in combination with regression techniques. It aims to identify the different existing methods and their characteristics.

RQ5: *What methods were proposed to achieve ensemble diversity?*
This question is focused on understanding the strategies and techniques used to ensure that the datasets (or models) within an ensemble are diverse. Diversity is crucial for an ensemble to have better performance,

as it combines different perspectives and minimizes the correlation between the errors of individual models.

RQ6: *What are the criteria and metrics used to evaluate the performance of dynamic selection techniques?*
This question seeks to understand the specific ways to measure the effectiveness and efficiency of dynamic selection techniques when applied to regression models. Accurate and comprehensive evaluation of these techniques is crucial to ensure that they improve performance compared to traditional methods.

These research questions provide a structured analytical focus for this systematic review, guiding the investigation towards specific gaps in the literature on DES applied to regression tasks. By systematically addressing these questions, this study aims to clarify existing evidence and highlight areas for further research.

After defining the research questions, the search strategy was detailed in the following subsection.

3.2.2. Search strategy

The objective of the search strategy was to find studies that would help answer the RQs described in subsections 3.1.1. The first phase of the strategy was to establish the search string. The second phase involved applying this search string to a selected set of digital libraries to find all primary studies. In the third phase, a search procedure was followed to ensure no relevant articles were left out. The three steps are described in detail in the following subsections.

3.2.3. Search string

The strings are constructed from the structure of the questions, and sometimes adaptations are necessary according to the specific needs of each database. Thus, the search strings were generated by combining key terms and synonyms using OR and AND, and possible peculiarities of the digital libraries, recording any adaptation whenever necessary. The search strings used for each database can be found in Table 2, and have been used on all fields.

We used the asterisk (*) wildcard to capture possible variations of terms, such as plurals and verb conjugations.

3.2.4. Literature resources

Relevant primary studies were identified through five recognized databases in the area of software engineering and computational intelligence (Table 3). These databases were chosen due to their relevance to the academic community.

The search string developed in Phase 1 was utilized in an automated search across all five databases to find journal and conference articles addressing the Research Questions outlined in Section 3.2.1. The search encompassed titles, abstracts, and keywords, with search terms adapted to each library's specific search engine requirements. The search strings used for each database can be found in Table 4.

Table 2
Search string.

((regress*) AND ("machine learning" OR "artificial intelligence") AND ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*"))

Table 3
Sources and digital libraries used.

Automatic search
ACM Digital Library (dl.acm.org);
IEEE Digital Library (IEEEXplore) (https://www.ieee.org);
Science Direct (Elsevier) (https://www.sciencedirect.com);
SCOPUS (https://www.scopus.com);
Web of Science (https://www.webofscience.com);

3.2.5. Search process

This phase aimed to locate articles relevant to the systematic literature review. The search process followed a two-stage approach:

Stage 1: Initial Search
Separate automated searches were conducted using the search string in each of the five databases. This initial search yielded a collection of potential primary studies.

Stage 2: Reference Check
The reference lists of relevant articles (primary studies meeting the inclusion and exclusion criteria defined in Section 3.3.1) were scrutinized to identify any studies potentially missed in Stage 1. Titles, abstracts, and keywords were used for initial assessment, with full-text examination employed when necessary. This step ensured comprehensive coverage of studies related to Dynamic Ensemble Selection.

With the planning and research questions established, the next phase involved conducting the review according to the protocol.

3.3. Conducting

This section details the process of conducting the SLR, which involved three key phases: Study Selection, Quality Assessment, and Data Extraction.

3.3.1. Study selection

After identifying potentially relevant studies through the search process (described earlier), we employed a rigorous two-step selection process to ensure the inclusion of the most relevant and reliable literature for our SLR.

Inclusion and exclusion criteria. Predefined inclusion (I) and exclusion (E) criteria (detailed in Tables 5 and 6) were applied to assess each study's suitability for the review.

Consensus-based evaluation. During the study selection phase, two reviewers jointly assessed each study through collaborative discussion. Rather than performing independent evaluations followed by a comparison step, all inclusion and exclusion decisions were determined immediately by consensus. As a result, no calculation of formal inter-rater reliability metrics, such as Cohen's Kappa (Cohen, 1960), was applicable. This collaborative approach ensured that judgments were consistently and transparently applied across all candidate studies.

3.3.2. Quality assessment

Following the application of inclusion criteria, selected studies underwent an additional Quality Assessment (QA) process. QA is crucial for data synthesis and interpreting results effectively and it typically relies on a checklist of criteria evaluated for each relevant study (Kitchenham & Charters, 2007).

We developed a quality assessment checklist based on the nine questions in Table 7. While the specific questions and scoring system for QA can be subjective, they provide a standardized framework to compare the selected articles. Each study received a quality score by summing the scores assigned to its answers for the quality control questions. Each question offered three possible answers: "Yes" (1 point), "Partially" (0.5 points), and "No" (0 points).

To ensure the review's findings' reliability, only relevant studies with an acceptable quality score threshold of 5 points were considered for subsequent data extraction and synthesis. Details of the quality scores for each study can be found in Table 8.

Quality assessment reliability. Similarly, during the quality assessment phase, both reviewers evaluated each study collaboratively, agreeing on the quality scores at the time of evaluation. Consequently, no formal inter-rater agreement measure such as Cohen's Kappa (Cohen, 1960) was applicable. The consensus-based scoring process ensured consistency and minimized subjective bias throughout the quality appraisal.

Table 4
Search string.

Database	Search
ACM	(Title:(regress*) OR Abstract:(regress*) OR Keyword:(regress*)) AND (Title:("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning") OR Abstract:("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning") OR Keyword:("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning")) AND (Title:("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*") OR Abstract:("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*") OR Keyword: ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*"))
IEEE	(("Abstract":regress*) AND ("Abstract":"machine learning" OR "Abstract":"artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("Abstract":"Dynamic Ensemble" OR "Abstract":"Dynamic Selection" OR "Abstract":"Dynamic Model Selection" OR "Abstract":"Dynamic Regress*")) OR (("Title":regress*) AND ("Title":"machine learning" OR "Title":"artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("Title":"Dynamic Ensemble" OR "Title":"Dynamic Selection" OR "Title":"Dynamic Model Selection" OR "Title":"Dynamic regression")) OR (("keyword":regress*) AND ("keyword":"machine learning" OR "keyword":"artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("keyword":"Dynamic Ensemble" OR "keyword":"Dynamic Selection" OR "keyword":"Dynamic Model Selection" OR "keyword":"Dynamic Regress*"))
Science Direct	(regression) AND ("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regression")
SCOPUS	TITLE-ABS-KEY ((regress* OR predict* OR estimat*) AND ("machine learning" OR "artificial intelligence") AND ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*"))
Web of Science	((TI = ((regress*) AND ("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*"))) OR AB = ((regress*) AND ("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regression"))) OR AK = ((regress*) AND ("machine learning" OR "artificial intelligence" OR "Neural Networks" OR "Deep Learning") AND ("Dynamic Ensemble" OR "Dynamic Selection" OR "Dynamic Model Selection" OR "Dynamic Regress*"))

Table 5
Inclusion criteria.

I1: Studies that answer one or more questions defined in this systematic review;
I2: Studies that include project estimation using hybrid models;
I3: Studies that use Dynamic Selection techniques;
I4: Studies that show which estimation techniques are most used and their respective areas.

Table 6
Exclusion criteria.

E1: Studies developed in languages other than English;
E2: Studies of type Editorial, poster, book, tutorial, and expanded abstract;
E3: Studies that are not related to the focus of this systematic review;
E4: Use the title, abstract and keywords as exclusion criteria for articles that are not related to the focus of the systematic review;
E5: Studies published before 2013
E6: Duplicate works;
E7: Studies not available for researcher access.

3.3.3. Data extraction

The Table 9 presents the data extraction form utilized to collect relevant information from each selected study to address the research questions of this review. The verified extracted data were documented in a file for use in the subsequent data synthesis stage. In Appendix A, we explicitly label each study with its respective IDs.

3.4. Reporting

The reporting phase synthesizes and organizes the extracted data to address the research questions comprehensively. By aggregating findings from the selected studies, we aim to provide a clear and structured overview of DES in regression tasks.

Table 7
Quality criteria.

QA1: Are the objectives presented in the study clearly defined?
QA2: Was the proposed solution clearly defined and discussed? Consider if the authors presented in the study: (i) Discussion of the proposal's specifics; (ii) Tables, charts, and statistical tests for comparing the results.
QA3: Does the study clearly describe the Dynamic Selection process (methodology)?
QA4: Was the experiment applied to a sufficient number of databases? More than 5 datasets - score (1.0); Between 3 and 5 datasets - score (0.5); Fewer than 3 datasets - score (0).
QA5: Does the study show which computational intelligence techniques were used, along with configuration parameters?
QA6: Was more than one evaluation metric used? 3 or more metrics interpreted - score (1.0); 2 metrics interpreted - score (0.5); 1 metric interpreted - score (0).
QA7: Was the proposed solution compared with the state-of-the-art? The number of techniques was considered, and if statistical tests were used. More than 1 technique and statistical tests - score (1.0); 1 technique with statistical tests - score (0).
QA8: Were the limitations, constraints, or threats to the validity of the results well discussed?
QA9: Are the study results clearly defined and discussed? We consider: (i) If the results were presented in tables, charts, and with statistical tests; (ii) If there is a section in the study that discusses the results; (iii) If the results were clearly discussed in the text.

3.4.1. Data synthesis

The main objective of data synthesis is to identify and group all evidence related to the RQs from the selected studies, as aggregation can facilitate and improve the generalization of the results. To enhance the presentation of these findings, we used visualization tools such as bar charts and tables. This synthesis is presented in Section 4.

4. Results and discussion

This section presents an analysis and discussion of the findings from the 21 primary studies reviewed, addressing each research question

Table 8
Evaluation of the quality of studies.

ID	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	Mean
PS01	1	0,5	1	1	0	1	0	0	1	5.5
PS02	1	1	1	1	1	1	0	0	1	7.0
PS03	1	1	0,5	1	0,5	0,5	1	0	1	6.5
PS04	0,5	1	1	1	0	1	0	0	1	5.5
PS05	1	1	1	0	0,5	1	0	0	1	5.5
PS06	1	0,5	1	1	0,5	0,5	0	0	1	5.5
PS07	1	1	1	1	1	1	1	0,5	1	8.5
PS08	1	1	0	1	1	1	1	0,5	1	7.5
PS09	1	1	1	1	1	1	0,5	0,5	1	8.0
PS10	1	1	1	0	1	0,5	1	0	1	6.5
PS11	1	1	1	0	1	0,5	0	0	1	5.5
PS12	1	1	1	1	0	1	0	0	1	6.0
PS13	1	1	1	1	1	1	1	0,5	1	8.5
PS14	1	1	1	1	1	1	1	1	1	9.0
PS15	1	1	1	0	1	0,5	0	0	1	5.5
PS16	1	1	1	1	1	1	1	1	1	9.0
PS17	1	1	1	0,5	1	1	1	0	1	7.5
PS18	1	1	0,5	0,5	1	0	1	0	1	6.0
PS19	1	1	0,5	1	1	0,5	1	0	1	7.0
PS20	1	1	1	0	0,5	1	0	0	1	5.5
PS21	1	1	1	1	1	0,5	1	0	1	7.5

Table 9
Information extracted from primary studies.

Information	Research question
Title	Overview
Author	Overview
Year of publication	Overview
Place of publication	Overview
Research method	Overview
Individual regression models used in the composition, combination of dynamic selection	RQ1
Metrics used to measure the competence region in the selection of regression models	RQ2
Data sets applied to dynamic selection with regression models	RQ3
Specific strategies, existing methods and their characteristics used to implement dynamic model selection	RQ4
Methods proposed to achieve ensemble diversity	RQ5
Metrics used to evaluate the performance of Dynamic Selection with regression models	RQ6

outlined earlier. We begin with an overview of the selected studies, followed by a detailed discussion of the review findings for each research question in separate subsections. Throughout the discussion, we interpret the review results both within the specific context of each research question and a broader context where relevant.

Although this section primarily adopts a structured and critical descriptive synthesis, a formal meta-analytic aggregation was not conducted. This decision was made due to the substantial heterogeneity in datasets, experimental protocols, and evaluation metrics across the selected studies, which would compromise the methodological validity of a statistical meta-analysis. Nevertheless, quantitative summaries and patterns are systematically highlighted to enhance analytical depth.

4.1. Overview of selected studies

Following the steps outlined in Sections 3.2.2–3.2.3, we collected the primary studies. Fig. 3 illustrates the number of studies gathered at each step across the different sources.

The automated search yielded 254 articles from five well-known online libraries: ACM, IEEE, Science Direct, Scopus, and Web of Science. After removing duplicates (101 articles) and applying exclusion criteria–such as publications before 2013 (30 articles), editorials, posters, books, tutorials, and extended abstracts (6 articles), and articles in languages other than English (2 articles)–115 articles remained for the pre-selection phase.

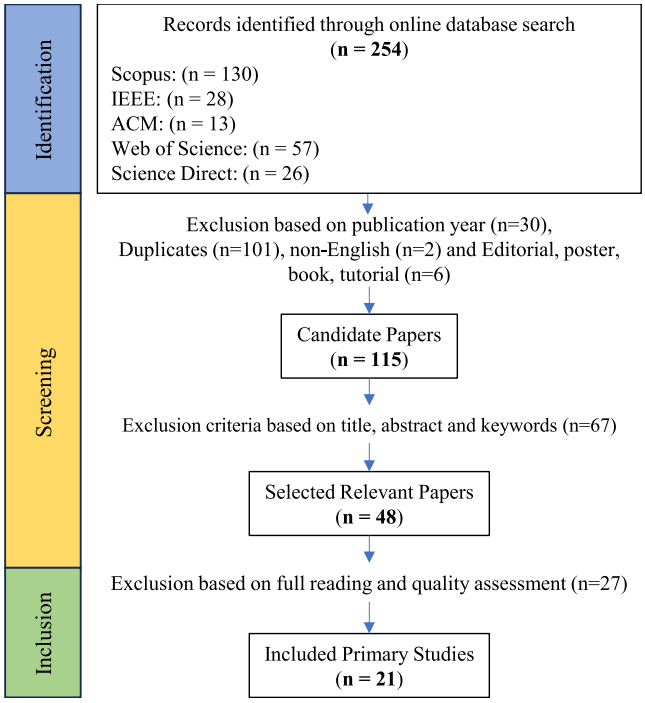


Fig. 3. Study selection process.

Table 10
Publication sources (journals and conferences).

Publication sources	Journals	Conference
ACM	1	0
IEEE	1	5
Science Direct	5	0
Scopus	8	1
Total	15 (71 %)	6 (29 %)

During the pre-selection phase, we applied additional exclusion criteria based on title, abstract, and keywords. This yielded a list of 48 articles deemed relevant or potentially relevant. Further application of inclusion and exclusion criteria, along with a quality assessment score threshold of 5.0, resulted in a final selection of 21 articles.

Table 10 shows the distribution of selected articles by publication source. Notably, 71 % of the selected studies were published in journals, while 29 % appeared in conference proceedings. Although we do not have empirical evidence, this higher percentage of journal publications may suggest an increased maturity in research on dynamic selection techniques.

An analysis of the publication trends over the years reveals a notable increase in studies on Dynamic Selection, particularly in 2021, 2022, and 2023, as shown in Fig. 4. Although the number of studies in 2024 is currently similar to previous years, it may increase as the year progresses.

This upward trend appears to be influenced by several converging factors. First, the growing availability of open-source ensemble learning libraries–such as scikit-learn, Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM)–has facilitated experimentation with complex model combination techniques, including dynamic selection. Second, the rising popularity of AutoML and meta-learning frameworks has promoted adaptive, instance-level model behavior, aligning closely with the principles of dynamic ensemble selection. Third, the increasing demands for predictive accuracy in domains characterized by nonstationarity and high data variability–such as energy forecasting, financial markets, and personalized healthcare–have motivated the development of locally adaptive regression models. These contextual and technological developments likely contributed to the

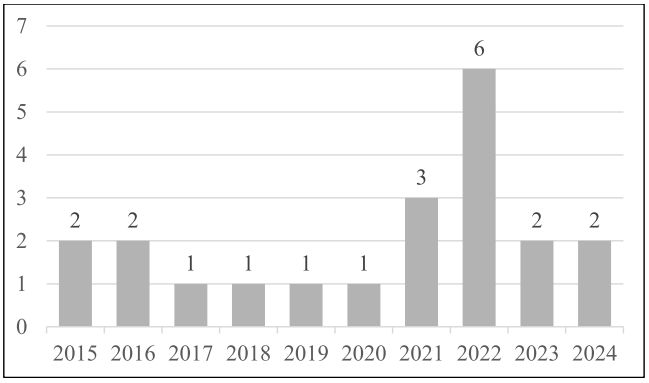


Fig. 4. Distribution of the 21 selected primary studies by year of publication (2015–2024).

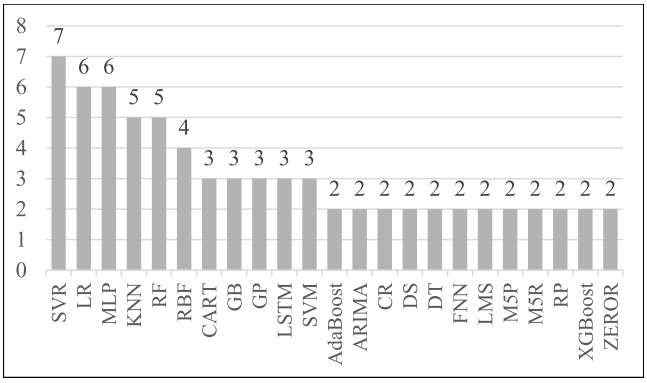


Fig. 5. Machine learning techniques used by studies.

Table 11
Quality levels of relevant studies.

Quality level	# Of studies	Percent
High (8.0 ≤ score < 9.0)	5	23.81
Medium (6.0 ≤ score < 8.0)	9	42.86
Low (5.5 ≤ score < 6)	7	33.33

recent surge in publications. While no causal relationship can be formally asserted, the correlation between these developments and the observed research output suggests a strong interplay between applied challenges and methodological advances.

In addition to the general increase in studies between 2021 and 2023, a more detailed examination of the selected articles reveals that this growth is not solely due to the broader popularity of machine learning or deep learning techniques. Rather, it reflects an expansion in the application domains and problem types addressed through dynamic selection strategies. In particular, recent studies have increasingly focused on areas such as healthcare (e.g., cancer survival prediction Xu et al., 2024, patient monitoring in decentralized systems Lee & Kang, 2024), energy forecasting (e.g., solar irradiance Sun et al., 2022, residential load prediction Paim & Enembreck, 2023), and financial market prediction (Moura et al., 2021b). Furthermore, methodological innovations, such as the integration of meta-learning approaches (Roy et al., 2016), dynamic weighting based on predictive uncertainty (Lee & Kang, 2024), and adaptive techniques for handling concept drift (Bardal et al., 2015), have contributed to the growing interest and diversification in research on dynamic ensemble selection for regression tasks. Therefore, the observed publication trend not only reflects the technological momentum of deep learning but also a significant broadening of methodological frameworks and application challenges, highlighting the maturity and evolution of this research area.

The quality of the selected studies was ensured by setting a minimum quality score threshold of 5 (out of a maximum of 9). As shown in Table 11, approximately 66.33 % (14 of 21) of the selected studies achieved a medium or high-quality level, suggesting that the final sample is reliable and of adequate quality for this review.

In the following subsections, we address each research question individually to provide a detailed analysis of the selected studies. By examining the findings for each research question, we aim to uncover patterns, identify common approaches, and highlight any gaps in the application of dynamic selection techniques to regression tasks. We begin with Research Question 1, which focuses on identifying the individual models used in dynamic selection frameworks.

4.2. RQ1: which individual models were used in the dynamic selection, composition, and combination?

The first research question aimed to identify which machine learning models were used as basic components in dynamic model selection. An analysis of the models and their frequency of use in the 21 primary studies revealed 59 distinct models. Among these, the majority (48 models – 81 %) were used only once or twice, indicating a remarkable diversity in the choices for constructing dynamic ensembles.

Fig. 5 illustrates the most frequently used models in the analyzed studies. Among the most commonly used models, three stand out: Support Vector Regression (SVR), Linear Regression (LR), and Multilayer Perceptron (MLP), cited in 7, 6, and 6 studies, respectively. The selection of these models can be attributed to their adaptive characteristics and generalization capabilities, which are essential attributes for dynamic selection frameworks.

The three most cited models–SVR, LR, and MLP–are widely employed in regression problems due to their effectiveness across different contexts. In the work of Cabral and Oliveira (2021), these three models were applied in the dynamic selection process for a software effort estimation problem. The authors proposed the PEETACO-DES method, which demonstrated superior performance over state-of-the-art techniques in Software Effort Estimation (SEE).

These algorithms were selected for their ability to handle complex and heterogeneous data, a common characteristic in effort estimation problems (Cabral & Oliveira, 2021). Linear Regression is valued for its simplicity and efficiency, especially in scenarios with linearly separable data and continuous data flow applications (Paim & Enembreck, 2023), while the MLP (Santos et al., 2022), with its neural network structure, captures nonlinear relationships and complex patterns. SVR (Liu & Zio, 2016), on the other hand, is known for its robustness in high-dimensional problems and its ability to handle noisy data, contributing to more stable predictions.

In addition, other regression models, such as the M5P algorithm (a model tree approach that integrates decision tree partitioning with linear regression models in the leaves), and K-Nearest Neighbors (KNN), were also used in Cabral and Oliveira's study (Cabral & Oliveira, 2021), promoting diversity and complementarity in predictions.

The preference for SVR and MLP across different studies can be attributed to their ability to handle complex, heterogeneous, and nonlinear patterns commonly found in real-world regression problems. SVR, by employing kernel functions, can map input data into higher-dimensional spaces, making it highly effective for problems with complex relationships and small to medium-sized datasets. Its inherent robustness to outliers and capacity for generalization also contribute to its frequent adoption in dynamic selection frameworks. Similarly, MLPs, with their layered neural network structures, offer flexibility in modeling non-linear and intricate relationships between variables, making them suitable for diverse domains ranging from time series forecasting to healthcare prediction. Moreover, MLPs benefit from recent advancements in training algorithms and regularization techniques, which have enhanced their stability and predictive performance even in noisy environments. This combination of generalization ability, adaptability, and scalability explains why SVR and MLP consistently

appear among the preferred base models in dynamic ensemble selection studies.

Random Forest (RF) and KNN were each used five times, reinforcing their role as popular choices for both regression and classification. These two models differ from more traditional ones, such as LR, MLP, and SVR, but share some similarities and show effectiveness in a wide variety of contexts. RF and KNN were applied together in two studies: Sani et al. (2022), which aimed to develop a dynamic regressor selection technique for trajectory loss prediction in multi-frequency and multi-environment contexts, and Cabral and Oliveira's study (Cabral & Oliveira, 2021), which proposed the PEETACO-DES method.

The diversity of models used demonstrates the flexibility and adaptability of dynamic selection frameworks, enabling the combination of algorithms with distinct characteristics to address complex problems. The works of Cabral and Oliveira (2021) and Sani et al. (2022) illustrate how the combination of models such as SVR, LR, MLP, RF, and KNN can lead to promising results in different domains, such as software effort estimation and trajectory loss prediction.

Over the years, the evolution has not only involved an increase in the number of models used but also a greater diversity and sophistication in the algorithms applied. This advancement is evidenced by both the increase in available models and the shift from simpler algorithms to more powerful and sophisticated methods. This trend reflects technological advancements and the growing complexity of the challenges researchers are attempting to address.

Diversity in dynamic ensembles plays a crucial role in enhancing adaptability and predictive accuracy, allowing the ensemble to respond more effectively to different patterns and variations in the data. By combining models with distinct characteristics, dynamic selection frameworks can leverage the strengths of individual algorithms, mitigating the weaknesses of some models in specific scenarios. For instance, while models like Linear Regression are efficient in capturing linear relationships, algorithms like MLP and SVR are more effective in nonlinear contexts and high-dimensional data. This variety enables the ensemble to select the most suitable model or combination for each data instance, maximizing the system's generalization capability and improving predictive accuracy (Cabral & Oliveira, 2021; Sani et al., 2022). Moreover, diversity reduces the risk of overfitting to a specific pattern, making the ensemble more robust and efficient across a wide range of applications.

The analysis of the models used in the 21 primary studies provided a comprehensive answer to the first research question, identifying the main components of dynamic ensembles. The prevalence of models such as SVR, LR, MLP, RF, and KNN suggests that these algorithms are well-suited to address the complexity and dynamics of the problems encountered in the analyzed studies.

4.3. RQ2: what metrics were used to measure the competency of the regression models in different regions of the data space?

After analyzing the regression techniques used in the studies (RQ1, detailed in Section 4.2), we then investigated the metrics employed to evaluate the performance of these models (RQ2). This analysis is crucial for understanding how researchers quantify prediction quality across different contexts.

In answering RQ2, we focused on how metrics are used to assess the competency of regression models in various regions of the data space, considering that the choice of an appropriate regression metric depends on multiple factors. These factors include the nature of the data, the specific objectives of the model, the presence of outliers, interpretability requirements, and the application context. There is no "perfect metric" applicable to all scenarios; the decision should be based on a clear understanding of the specific characteristics of the regression problem (Chai & Draxler, 2014; Kuhn & Johnson, 2007).

The selected studies revealed the use of 26 different metrics for model validation. In the context of regression models, it is essential to understand which metrics are most frequently employed to assess model

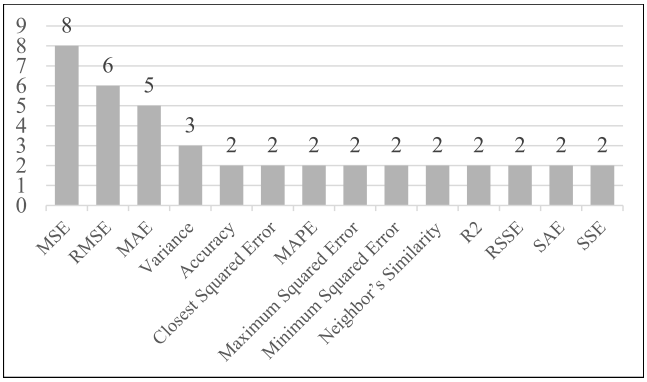


Fig. 6. Metrics used to measure the competency of the regression models.

competency across different data regions. This high number of metrics reflects the diversity of approaches in measuring regression model performance and the complexity involved in selecting the most appropriate metric for varied data scenarios. To visualize the distribution of metrics used in the studies, we present Fig. 6, which shows metrics with two or more occurrences.

Among the 26 identified metrics, Mean Square Error (MSE) is the most commonly used, with 8 occurrences. MSE is calculated as the mean of the squared differences between predicted and actual values, penalizing larger errors more strongly. This characteristic makes MSE sensitive to outliers, as larger errors significantly increase its final value.

Next, we have Root Mean Squared Error (RMSE), with 6 occurrences, which is the square root of MSE, bringing the error back to the same scale as the input variables. Mean Absolute Error (MAE), appearing 5 times, is the mean of the absolute differences between actual and predicted values. Unlike MSE and RMSE, MAE does not intensify larger errors, making it less sensitive to outliers.

Variance (3 occurrences) is a dispersion measure that indicates how much the values of a variable deviate from the mean. In modeling, variance can also represent how much model predictions vary. In machine learning, high variance suggests that the model is sensitive to fluctuations in training data (overfitting), while low variance may indicate that the model is more consistent but potentially underfitted.

The reviewed studies highlight three primary error metrics for evaluating regression model competency: Mean Squared Error, Root Mean Squared Error, and Mean Absolute Error, each applied in specific contexts and with characteristics that complement different evaluation needs.

MSE is widely used for its ability to heavily penalize large deviations, making it ideal for scenarios where accuracy in capturing large variations is essential. For example, in time series forecasts, MSE facilitates the comparison between regressor selection methods, allowing for continuous analysis of model accuracy over time (Moura et al., 2021b). In environmental issues, such as particulate matter forecasting, MSE provides a consistent measure of model effectiveness in capturing particle concentration variability (Bueno et al., 2020). This metric has also proven effective in solar power generation forecasting (Sun et al., 2022), in hybrid systems for continuous forecasting (Oliveira et al., 2022), and in dynamic data flows with concept drift, adapting to varied application contexts (Liu & Zio, 2016; Paim & Enembreck, 2023).

Complementing MSE, RMSE is widely applied for returning error values to the same scale as the input variables, facilitating interpretation. It has been used, for example, in trajectory loss prediction in multi-frequency environments, being sensitive to context variations and different environmental conditions (Sani et al., 2022). In healthcare, RMSE has been applied to patient survival time forecasts, where variations in health conditions directly affect model accuracy (Xu et al., 2024). In renewable energy, it was essential for improving the reliability of solar irradiance predictions (Santos et al., 2022). In concept drift

and meta-learning studies, RMSE helped evaluate and adjust models to recurring changes and short-term residential demand forecasting (Bardal et al., 2015; Roy et al., 2016; Yang et al., 2023).

Finally, MAE stands out for its simplicity and lower sensitivity to outliers, making it ideal for a wide range of forecasting problems where robustness and precision are required. In the study of signal loss in multi-frequency environments, MAE quantified the ensemble's performance in different trajectory conditions (Sani et al., 2022). In medical applications, such as predicting gastric cancer patient survival time, MAE improved accuracy in survival estimates (Xu et al., 2024). In solar and residential load forecasting, MAE validated model reliability in response to irradiance and consumption variations (Sun et al., 2022). In decentralized neural networks, MAE enabled adjustments in dynamic models based on predictive uncertainty, proving effective in distributed learning scenarios (Lee & Kang, 2024).

In summary, the diversity of identified metrics–including MSE, RMSE, and MAE–reflects the necessary adaptability for dynamic model selection approaches to capture data complexities and variations across multiple contexts. Each metric has characteristics that serve different modeling objectives, from penalizing large errors with MSE to the outlier robustness offered by MAE. Metric selection is, therefore, a strategic decision based on the specificities of each regression problem, ensuring that evaluations are adapted to the needs of each application. Consequently, the analysis of metrics reinforces the importance of a careful approach in validating dy namic regression models, essential for ensuring accurate and reliable predictions across a wide range of data scenarios.

To enhance clarity regarding the models employed, the evaluation metrics adopted, and the application domains addressed by each primary study, Table 12 provides a consolidated overview.

To facilitate its interpretation, the following acronyms refer to common regression and ensemble models, as well as evaluation metrics: CART (Classification and Regression Trees), AdaBoost (Adaptive Boosting), XGBoost (Extreme Gradient Boosting), LightGBM (Light Gradient Boosting Machine), LSTM (Long Short-Term Memory), CNN (Convolutional Neural Network), FNN (Feedforward Neural Network), LOF (Local Outlier Factor), OCSVM (One-Class Support Vector Machine), ELM (Extreme Learning Machine), GBM (Gradient Boosting Machine), ARIMA (AutoRegressive Integrated Moving Average), and Meta-DES (Meta-learning-based Dynamic Ensemble Selection). The acronym SFNR refers to a specific ensemble strategy proposed in the corresponding study (PS14).

In summary, the choice of models such as SVR, MLP, and RF can be explained by their recognized ability to handle heterogeneous data distributions, capture complex patterns, and provide stable generalization across different domains. SVR, with its capability to handle high-dimensional and noisy datasets, has been a frequent choice for software effort estimation and solar energy forecasting (Cabral & Oliveira, 2021; Santos et al., 2022). MLPs, due to their nonlinear approximation power, have been effectively applied in financial forecasting and environmental modeling (Bueno et al., 2020; Moura et al., 2021b). Random Forest, with its ensemble-based structure and robustness against overfitting, has proven effective in contexts involving noise and feature redundancy, such as wireless communication prediction and outlier detection in industrial settings (Sani et al., 2022; Wang et al., 2022). These characteristics justify the frequent adoption of these models among the reviewed studies and highlight their relevance in dynamic selection scenarios requiring adaptability, robustness, and predictive performance.

4.4. RQ3: what datasets were used and how were they applied in the dynamic selection of regression models?

Dynamic selection of regression models has been explored using a wide variety of datasets, showcasing the versatility and applicability of these approaches across different domains (Table 13).

In analyzing the selected studies, 144 unique datasets were identified, spanning real-world and synthetic data from diverse contexts such as environmental monitoring, healthcare, finance, industrial processes, and software engineering. This dataset variety highlights the broad applicability of dynamic methods and their capacity to adapt to distinct challenges.

4.4.1. Environmental and synthetic datasets

Environmental datasets, such as PM10 Concentration (Bueno et al., 2020), collected from Brazilian cities like Campinas, Jundiaí, and São Caetano do Sul, were used to predict air quality. These datasets, with hourly measurements of PM10 concentrations over several years, tested the ability of dynamic ensemble methods to handle highly variable data.

Complementing real-world datasets, synthetic datasets like Friedman Artificial and Hyperplane Artificial (Bueno et al., 2020) were instrumental in assessing algorithm robustness and generalization under controlled conditions. These benchmarks allowed researchers to refine their strategies and validate performance under standardized scenarios.

4.4.2. Energy and healthcare datasets

Energy-related datasets, such as Energy Efficiency and Cycle Power Plant (Kurzynski et al., 2018), served as benchmarks to evaluate dynamic ensemble methods in forecasting scenarios. These datasets encapsulate challenges related to energy systems modeling, providing insights into variability and predictive accuracy in industrial applications.

In healthcare, datasets like Breast, Liver, and Blood (Roy et al., 2016) were among the 30 binary datasets used to evaluate the META-DES framework ability to dynamically select competent models. Specifically, the Breast Cancer dataset, comprising 194 instances and 32 attributes, showcased the effectiveness of ensemble diversity in enhancing predictive performance across healthcare applications (Moura et al., 2019, 2021a).

4.4.3. Financial and solar energy datasets

Financial datasets, including Amazon, Apple, and Goldman, highlighted the adaptability of dynamic methods in time series forecasting (Moura et al., 2021b). These datasets demonstrated the potential of techniques like Dynamic Weighting (DW) in addressing high variability in financial markets.

The DKASC Solar Centre dataset (Sun et al., 2022), containing two years of solar energy generation data, illustrated the effectiveness of dynamic regression models in adapting to temporal variations, enabling one-hour-ahead energy generation forecasting.

4.4.4. Software engineering datasets

Software development effort estimation, such as Cocomo81, Cocomonasa, ISBSG, Maxwell, and Desharnais, were applied to predict software development effort (Cabral et al., 2017; Cabral & Oliveira, 2021). Smaller datasets, with fewer than 100 projects, the leave-one-out technique was used., while larger datasets were split into training (70 %) and testing (30 %) subsets. The results indicated that dynamic selection of regression algorithms improved productivity and cost estimates, consistently outperforming other approaches in statistical evaluations.

The diverse application of these datasets underscores the flexibility of dynamic regression models to address a wide range of challenges. Whether dealing with complex real-world variability, such as air quality and healthcare data, or leveraging controlled synthetic environments for algorithm benchmarking, these datasets provided a robust framework for validating and improving dynamic selection strategies.

In conclusion, the wide array of datasets utilized across the reviewed studies highlights the essential role of diversity in advancing dynamic regression methods. By addressing challenges in varied domains–from environmental monitoring to finance and healthcare–these datasets have paved the way for more adaptable, precise, and reliable predictive models. Such diversity ensures that dynamic methods are not only theoretically robust but also practically applicable to real-world problems, reinforcing their relevance in tackling complex data scenarios.

Table 12
Summary of models, metrics, and domains by study.

Study ID	Models used	Metrics used	Domain
PS01	Linear Regression, SVR, Random Forest	RMSE, MSE	Energy Efficiency Dataset
PS02	MLP, SVR, Bagging Regressor	RMSE, MAE, R^2	Finance (Stock Market Prediction)
PS03	Random Forest, MLP, SVR, CART	MAE, MSE, RMSE, R^2	Environmental Monitoring (PM10, Synthetic Datasets)
PS04	KNN, Random Forest, Bagging	MAE, RMSE, R^2	Wireless Networks (Signal Loss)
PS05	Random Forest, AdaBoost	MAE, RMSE	Traffic Accident Severity
PS06	Meta-DES, kNN Pool	MAE, MSE, RMSE	Healthcare (Breast, Liver, Blood Datasets)
PS07	MLP, SVR, RF, KNN	RMSE, MAE	Software Effort Estimation
PS08	XGBoost, SVR, RF	RMSE, MAE, R^2	Healthcare (Cancer Survival)
PS09	MLP, SVR, LSTM, CNN	RMSE, MAE, MAPE	Time Series Forecasting
PS10	Isolation Forest, LOF, OCSVM	MSE, RMSE	Industrial Process (Steel Temperature Prediction)
PS11	SVR, LR, LSTM, FNN	MAE, RMSE	Solar Energy Forecasting
PS12	Meta-DES, Pool-based Selection	RMSE, MAE	Data Streams
PS13	SVR, RF, MLP, KNN	MAE, RMSE, MAPE	Software Effort Estimation
PS14	Meta-DES, SFNR Ensemble	RMSE, MAE	Concept Drift Detection
PS15	SVR, MLP, Bagging, Boosting	MAE, RMSE	Wireless Networks
PS16	Pool of 10 models	MAE, RMSE, MAPE	Various Regression Benchmarks
PS17	SVR, ELM, RF, GBM, ARIMA	RMSE, MAE, MAPE	Solar Energy Forecasting
PS18	Ensemble of Linear Models	RMSE, MAE	Concept Drift (Data Streams)
PS19	Adaptive Linear Regression	RMSE, MAE, MAPE	Short-Term Load Forecasting
PS20	Dynamic Weighted Ensemble	RMSE, MAE	Time Series Forecasting
PS21	Decentralized Neural Networks	RMSE, MAE	Healthcare and Distributed Systems

Table 13
Number of datasets used by studies selected in our SLR.

ID	Total	ID	Total	ID	Total
PS01	25	PS08	1	PS15	1
PS02	10	PS09	9	PS16	15
PS03	11	PS10	1	PS17	4
PS04	2	PS11	1	PS18	3
PS05	1	PS12	1	PS19	5
PS06	30	PS13	16	PS20	2
PS07	8	PS14	20	PS21	9

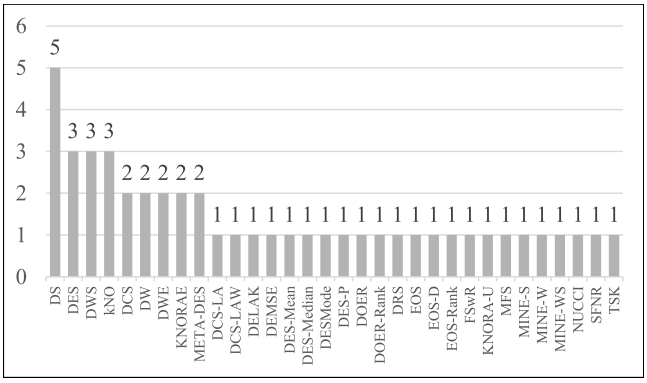


Fig. 7. Different types of dynamic selection techniques.

4.5. RQ4: what are the different types of dynamic selection techniques used in conjunction with regression methods?

Dynamic selection techniques, when applied to regression methods, have demonstrated versatility in adapting to varying data characteristics and improving predictive accuracy (Moura et al., 2021a). Various methods and strategies are used to implement dynamic selection, as shown in Fig. 7, with different approaches and evaluation criteria, such as Dynamic Selection (DS) Moura et al. (2021b), Dynamic Ensemble Selection (DES) Sani et al. (2022), k-Nearest Output Profiles (kNO) Oliveira et al. (2022), Dynamic Weighting (DW) Moura et al. (2019), Dynamic Weighted Selection (DWS) Moura et al. (2019), and Meta-learning-based Dynamic Ensemble Selection (META-DES) Khattak et al. (2022), among others. This section explores the strategies and methods identified in the reviewed studies, categorizing their approaches and highlighting their characteristics, applications, and limitations.

Among the analyzed studies, Moura et al. (2021b) and Moura et al. (2019) emphasize the importance of defining regions of competence to dynamically select the most suitable models. Methods such as Dynamic Selection, Dynamic Weighting, and Dynamic Weighting with Selection use criteria such as accumulated error, variance, and similarity to determine model competence within these regions. The results indicate that dynamic model combinations outperform single-model performance, as well as averaging or median-based approaches, although the choice of the ideal competence measure depends on the problem and the analyzed context.

Conversely, Cabral et al. (2017) present a heterogeneous approach employing various regression algorithms. In this approach, classifiers dynamically select the most appropriate model for each instance to estimate software development effort. The results show that dynamic selection, especially when based on multiple criteria, outperforms static or homogeneous methods in terms of accuracy.

Yang et al. (2023) propose a dynamic method that adjusts model weights in real time, using state-space models and particle filters to enhance the robustness and precision of the ensemble. Complementarily, Lee and Kang (2024) utilize predictive uncertainties, estimated via Monte Carlo Dropout, to dynamically weigh neural networks trained in decentralized environments, ensuring better generalization in scenarios where data is distributed and not shareable. These approaches illustrate the diversity of employed techniques and demonstrate how different strategies can be adapted to specific contexts, such as time series forecasting, electric load prediction, and decentralized environments.

Sani et al. (2022) propose a dynamic selection technique for regression and ensembles aimed at predicting signal loss in wireless networks across multiple frequencies and environments. Using clusters defined by the k-means algorithm, the most suitable models are selected based on the minimum absolute error, demonstrating accuracy gains in imbalanced data scenarios. Wang et al. (2022) explore steel temperature prediction in industrial furnaces using a dynamic ensemble for outlier detection, combining regression-derived information with clusters to define the region of competence and measure detector competence. Liu and Zio (2016), in turn, present a dynamic ensemble based on Support Vector Regression (SVR) for data streams with recurring patterns, utilizing dynamic selection of relevant submodels for each new data point. This strategy reduces computational complexity and improves adaptation to non-stationary patterns.

Oliveira et al. (2022) propose a hybrid system for time series forecasting, combining linear and non-linear models through residual forecasting. Using the Dynamic Selection algorithm, the method identifies the most suitable model for specific residual patterns in a heterogeneous set including MLP, SVR, LSTM (Long Short-Term Memory network), CNN (Convolutional Neural Network), among others. Sun et al. (2022) develop a dynamic framework for solar energy generation forecasting, where base model weights are dynamically adjusted based on local performance, using SVR, LR, FNN (Feedforward Neural Network), and LSTM as base learners. Santos et al. (2022) present a heterogeneous dynamic selection model (HetDS) for solar irradiance forecasting, employing ARIMA, SVR, MLP, ELM, DBN (Deep Belief Network), RF, and GBM (Gradient Boosting Machine) in a model pool. Selection is performed locally based on accuracy in regions of competence, optimizing performance for specific patterns. These studies highlight the application of dynamic selection to address diverse and dynamic patterns in time series, demonstrating gains in accuracy and robustness.

Finally, Khattak et al. (2022) propose an interpretable dynamic selection method to predict the severity of traffic accidents, using algorithms such as META-DES, KNORAE (k-Nearest Oracles Eliminate), and DES-P (Dynamic Ensemble Selection using Performance), with models like Random Forest (RF) and AdaBoost (Adaptive Boosting). The approach incorporates SHAP explanations, highlighting risk factors such as driver age and weather conditions. Roy et al. (2016), in turn, use meta-learning to recommend optimal pool sizes for classifiers in META-DES systems, employing complexity measures to optimize computational efficiency and accuracy. These studies underscore the adaptability of dynamic selection techniques to various challenges, prioritizing interpretability and efficiency.

In summary, dynamic selection techniques for regression methods demonstrate broad applicability across diverse contexts. From defining regions of competence and selection criteria to incorporating adaptive and explanatory strategies, these approaches emphasize their importance in enhancing predictive models, especially in scenarios with dynamic patterns and heterogeneous characteristics.

4.6. RQ5: what methods were proposed to achieve ensemble diversity?

The diversity within model ensembles is crucial for enhancing robustness and accuracy in machine learning tasks. Various methods have been proposed in the literature to achieve this diversity, particularly in dynamic selection systems.

One of the most common approaches is the use of heterogeneous ensembles, which combine different learning algorithms to leverage the complementary characteristics of models. Examples include combinations of neural networks, decision trees, and Support Vector Machines (SVR), resulting in increased robustness and accuracy (Cabral et al., 2017; Mendes-Moreira et al., 2015; Oliveira et al., 2022; Sani et al., 2022). This heterogeneity can also be enhanced by varying the parameters of algorithms to broaden diversity (Liu & Zio, 2016; Mendes-Moreira et al., 2015).

Bagging and random sampling techniques are widely recognized as effective methods for promoting diversity in ensembles (Cabral & Oliveira, 2021; Moura et al., 2021b; Paim & Enembreck, 2023; Wang et al., 2022). Bagging employs random subsets of training data, often generated via bootstrap, to construct multiple models, reducing correlation between them and improving the ensemble's generalization (Wang et al., 2022). For instance, bootstrap with replacement has been used to train individual models in homogeneous ensembles, ensuring different perspectives on the data (Moura et al., 2021b). In a data stream context, bagging with Poisson distribution ($\lambda = 6$) increased diversity by adjusting instance weights in each training subset, demonstrating flexibility for different objectives (Paim & Enembreck, 2023). Additionally, bagging applied to KNN regressors in wireless network domains proved effective in reducing model correlation and improving predictive accuracy, outperforming techniques like Random Forest (Cabral & Oliveira, 2021).

Feature bagging, on the other hand, randomly selects attribute subsets to train each model, promoting ensemble diversity by forcing models to focus on different aspects of the data. This technique has been successfully applied in outlier detection ensembles, where randomness in attribute selection enhances the ensemble's ability to capture varied patterns in the data (Wang et al., 2022). Together, these approaches demonstrate how bagging and feature bagging can create more robust, accurate, and adaptable ensembles for diverse scenarios and problems.

Clustering-based methods also promote diversity by dividing data into competence regions, enabling models to specialize locally (Sani et al., 2022; Xu et al., 2024). In (Xu et al., 2024), clustering was used to group patients with similar survival profiles and train base regressors specialized in these groups. Algorithms like weighted k-means and fuzzy k-means are employed to create clusters. The similarity between a new patient and clusters is then used to calculate the weights of base regressors during prediction, dynamically adjusting weights based on the model's local performance.

Adaptive sliding window techniques in continuous data streams offer an innovative approach to handling concept drift, where data distributions evolve over time, challenging traditional machine learning methods (Bueno et al., 2020; Paim & Enembreck, 2023). This strategy dynamically adjusts window sizes used to train ensemble models, ensuring recent patterns are prioritized while retaining knowledge of previous concepts. For example, in Bueno et al. (2020), the EOS (Dynamic and Online Ensemble for Regression) framework manages a pool of models by periodically including new regressors and removing outdated ones, promoting long-term memory and structural adaptability. Within this framework, the DOER (Dynamic and Online Ensemble Regressor) component dynamically adjusts the weights of base models based on their recent predictive performance, enabling fine-grained local adaptation to changing data patterns. In (Paim & Enembreck, 2023), Adaptive Linear Regression (ALR) exploits temporal diversity by training linear regressors on windows of different sizes, balancing sensitivity to recent changes with noise stability. This approach addresses the "elasticity-plasticity" dilemma, where smaller windows adapt quickly but are more susceptible to noise, while larger windows offer robustness but may be slower to capture new trends. These methods demonstrate how prioritizing recent patterns and temporal diversification can enhance ensemble adaptability and accuracy in dynamic data streams.

Additionally, concept drift detection allows ensembles to identify changes in data distributions, while the inclusion of new models enables adaptation to new patterns, maintaining the ensemble's diversity and generalization capability (Barddal et al., 2015; Liu & Zio, 2016). The Online Ensemble based on Feature Vectors (OE-FV) proposed in (Liu & Zio, 2016) incorporates the inclusion of new submodels as a strategy for handling pattern drift. When a new pattern is detected and no existing submodel adequately represents it, OE-FV creates and adds a new submodel to the ensemble. This dynamic inclusion of submodels allows OE-FV to adapt to new concepts and maintain its accuracy in dynamic environments. In (Barddal et al., 2015), a social network-based approach for concept drift detection in regression tasks is presented. The Scale-free Network Regressor (SFNR) algorithm maintains a dynamic ensemble of regressors. SFNR uses the Adaptive Window (ADWIN) algorithm to detect concept drifts, monitoring base model error over time and signaling significant changes in data distributions. When a drift is detected, SFNR adjusts the ensemble by adding or removing models to adapt to the new concept.

Hybrid systems also stand out, combining linear and non-linear models to address distinct data patterns (Kurzynski et al., 2018). For instance, the MINE (Meta Integration) framework for dynamic regressor selection can be applied to homogeneous or heterogeneous ensembles. Although the article does not focus specifically on hybrid systems, it highlights the importance of using different machine learning

algorithms to construct more robust and accurate ensembles (Moura et al., 2021a). Combining linear and non-linear models in the ensemble can be an effective strategy for addressing distinct data patterns. The study (Oliveira et al., 2022) proposes a hybrid system called DReF (Dynamic Residual Forecasting) for time series forecasting. DReF combines a linear statistical model (ARIMA or SARIMA) for linear forecasts with a pool of non-linear machine learning models to model residuals (errors) from linear forecasts. The study argues that real-world time series often comprise both linear and non-linear patterns, making hybrid systems more suitable for handling this complexity. DReF uses a dynamic selection algorithm to choose the most appropriate non-linear model for residual prediction, leveraging the strengths of different non-linear models to capture specific patterns.

Additionally, diversity can be achieved through dynamic selection strategies such as Dynamic Weighting (DW) and Dynamic Weighting with Selection (DWS) Cabral and Oliveira (2021); Kurzynski et al. (2018); Mendes-Moreira et al. (2015); Moura et al. (2021a,b); Oliveira et al. (2022). These techniques continuously evaluate model competence in specific data regions, dynamically selecting those with better local performance and reducing the impact of less effective models. This process allows the ensemble to adapt to local data characteristics, prioritizing the most competent models in specific contexts. The use of DW and DWS contributes to the creation of more robust, flexible, and accurate ensembles, capable of handling variability and complexity across different prediction problems, offering improved performance in tasks requiring high adaptability.

Finally, strategies based on social network theories have been employed to manage dynamic ensembles, such as using the scale-free network model to organize and update models based on their relevance within the network. This approach identifies and prioritizes more competent models, reducing the impact of less effective ones, resulting in a more robust and accurate system for addressing concept drift in data streams (Barddal et al., 2015).

In summary, methods to promote diversity in ensembles encompass a wide range of strategies, from model heterogeneity to adaptive data-driven techniques. The choice of the most suitable approach depends on the context and data characteristics, but the common goal is always to maximize the robustness and accuracy of ensembles, fully exploiting the different perspectives models can offer.

4.7. RQ6: what are the criteria and metrics used to evaluate the performance of dynamic selection techniques?

Dynamic selection techniques aim to identify the most competent models from a set for each test instance, focusing on maximizing prediction accuracy and adapting to data characteristics. Their performance evaluation is structured around criteria that define the most relevant aspects to be analyzed and the metrics that quantify these criteria.

Evaluation criteria play a crucial role in analyzing the performance of dynamic selection techniques by prioritizing key aspects for examination during experiments (Moura et al., 2021b; Sani et al., 2022). Among the most frequently reported criteria are prediction accuracy, robustness in adverse scenarios, computational efficiency, interpretability, and scalability.

Prediction accuracy is the most commonly addressed criterion, present in all the analyzed studies. It assesses the technique's ability to minimize prediction errors, typically measured by metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE), among others. The choice of metric depends on the specific context of the problem and the relative importance of larger or smaller errors.

Robustness refers to the ability of techniques to maintain consistent performance in the face of challenges such as noise and data imbalance. Studies such as Sani et al. (2022); Xu et al. (2024) have demonstrated significant improvements in adverse scenarios, such as predictions with imbalanced datasets or outliers.

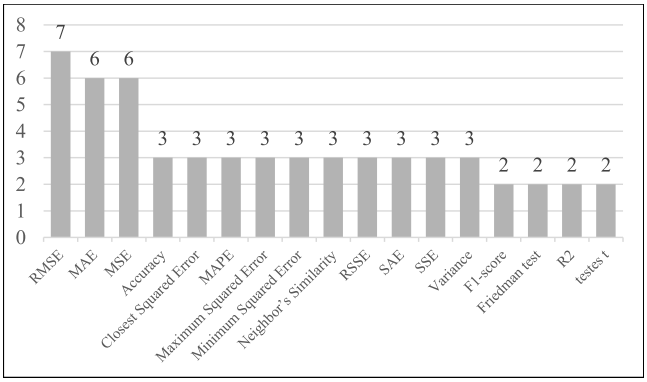


Fig. 8. Metrics used in ensemble performance validation.

Computational efficiency evaluates the execution time and resources required to implement the techniques, especially for large datasets or continuous data streams. While most articles do not directly address computational efficiency or provide performance metrics, nearly half of the studies highlight its importance. For example, in Roy et al. (2016), the authors emphasize reducing computational costs by predicting the optimal pool size of classifiers, avoiding brute force methods. Efficiency was central to the META-DES framework, where selecting an appropriate pool size significantly reduced execution time. Similarly, Barddal et al. (2015) presented computational efficiency metrics, evaluating average execution time under various scenarios and comparing it with other methods like FIMT-DD (Fast Incremental Model Tree with Drift Detection) and AMRules (Adaptive Model Rules).

Interpretability and explainability focus on the ability to explain the decisions made by ensembles. Techniques like SHAP (SHapley Additive exPlanations) enhance decision transparency and reliability for end-users (Khattak et al., 2022). SHAP analysis has been used to identify the most relevant risk factors for injury severity (e.g., driver age, day of the week) and explain each factor's contribution to predictions, transforming "black-box" models into more comprehensible approaches.

Finally, scalability assesses how a technique performs in contexts with a significant increase in data volume or problem complexity. This is particularly relevant for continuous data streams, where models must efficiently handle large volumes and constant changes (Paim & Enembreck, 2023). Scalability ensures that methods can be applied in large-scale real-world scenarios, as demonstrated by a framework evaluation on public datasets of varying sizes, including a set with 15,000 samples, without compromising predictive performance (Xu et al., 2024).

Although the studies address these criteria in various ways, there is a notable lack of research that simultaneously considers multiple criteria. This highlights the need for more integrated analyses in the development of new approaches. Table 14 presents a matrix summarizing the evaluation criteria, representing the qualitative aspects guiding the analysis of dynamic selection techniques' performance in the selected studies.

To measure these criteria, researchers use various specific metrics, including Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), accuracy, and the coefficient of determination (R^2). These metrics quantify the effectiveness of the techniques across different dimensions of performance (Fig. 8).

RMSE was the most widely used metric among the analyzed studies, appearing in seven instances. By penalizing larger deviations between predicted and actual values, RMSE is particularly suitable for scenarios where precision in predictions with high variability is essential.

For example, in predicting signal loss in wireless networks, RMSE was the primary metric used to compare the accuracy of dynamic and static models. The dynamic selection technique outperformed traditional methods, especially in imbalanced data environments. Additional

Table 14
Matrix of evaluation criteria guiding the analysis of dynamic selection techniques' performance.

ID	Prediction accuracy	Robustness	Computational efficiency	Interpretability	Scalability
PS01	True	False	False	False	False
PS02	True	False	False	False	False
PS03	True	True	True	False	False
PS04	True	True	False	False	False
PS05	True	False	False	True	False
PS06	True	False	True	False	False
PS07	True	False	False	False	False
PS08	True	True	False	False	True
PS09	True	True	True	False	False
PS10	True	True	True	False	False
PS11	True	False	False	False	False
PS12	True	True	True	False	False
PS13	True	True	False	False	False
PS14	True	True	False	False	False
PS15	True	True	False	False	False
PS16	True	True	False	False	False
PS17	True	False	False	False	False
PS18	True	False	True	False	False
PS19	True	False	True	False	True
PS20	True	True	True	False	False
PS21	True	False	True	False	False

metrics, such as MAE, MAPE, and R^2, were also employed to evaluate performance, reinforcing the approach's effectiveness in minority regions of the data space, where static models exhibited higher error rates (Sani et al., 2022).

Similarly, for predicting survival time in gastric cancer patients, RMSE, MAE, and R^2 validated the effectiveness of a dynamic ensemble framework based on weights. This framework achieved improvements of over 13 % compared to models like GBR (Gradient Boosting Regressor), DELAK (Deep Ensemble Learning with Attention and Knowledge), and DWE (Dynamic Weighted Ensemble), emphasizing its capacity to personalize predictions and enhance clinical decision-making support in complex scenarios (Xu et al., 2024).

In short-term electrical load forecasting, RMSE, combined with MAE and MAPE, demonstrated how real-time adjusted dynamic models significantly improved accuracy in contexts with high data variability. This combination of metrics was crucial for making more assertive decisions in energy management and resource allocation (Yang et al., 2023).

Including MAE, which appeared in six instances, allows for understanding both the average magnitude of errors (MAE) and the presence of large errors and outliers (RMSE), contributing to a more detailed and precise evaluation of dynamic ensemble techniques.

Beyond traditional error metrics, some studies utilized R^2, which assesses the proportion of variance explained by the ensemble. This metric demonstrates the selection mechanisms' efficacy in capturing data variability (Sani et al., 2022; Xu et al., 2024).

The diverse range of criteria and metrics analyzed across the studies highlights the flexibility of dynamic selection techniques in meeting the demands of different scenarios and domains. From precision in imbalanced data to explainability and scalability in continuous streams, these approaches offer robust and adaptable solutions. Future research should integrate multiple criteria into comprehensive frameworks, ensuring these techniques evolve to address the growing challenges of complex systems.

Although computational efficiency was not the primary focus in most of the selected studies, some works (e.g., Barddal et al., 2015; Roy et al., 2016) emphasized its importance when evaluating dynamic selection techniques. In particular, Roy et al. (2016) optimized the classifier pool size to reduce unnecessary computational effort, while Barddal et al. (2015) assessed execution times under various drift scenarios. Across the reviewed literature, a recurring trade-off is observed: dynamic selection methods tend to increase computational costs due to instance-wise model evaluation, but this overhead is often compensated by improved predictive performance and adaptability. Therefore, the choice to adopt dynamic selection strategies should consider the criticality of execution time and resource availability in each application context, especially in real-time systems where latency constraints are stringent.

4.8. Critical comparison and methodological synthesis

Beyond describing the features of the selected studies, a critical synthesis was conducted to assess their methodological rigor and overall contributions. The studies varied considerably in terms of experimental design, dataset diversity, evaluation criteria, and methodological robustness. While all studies addressed prediction accuracy, approximately 76 % also incorporated additional aspects such as robustness under adverse conditions, computational efficiency, interpretability, or scalability (Table 14). Studies like Xu et al. (2024) and Yang et al. (2023) demonstrated stronger methodological frameworks by integrating multiple evaluation criteria, while others focused predominantly on predictive performance.

In terms of datasets, studies that employed a broad range of real-world and synthetic datasets (e.g., Moura et al., 2019; Roy et al., 2016) were considered more robust, allowing broader generalization of findings. Regarding algorithmic diversity, works such as Oliveira et al. (2022) and Cabral and Oliveira (2021) were distinguished by applying heterogeneous ensembles and exploring dynamic weighting strategies across different domains. However, a limitation noted across many studies was the lack of systematic computational efficiency analysis, with only a few explicitly reporting runtime metrics or resource usage. Overall, the reviewed literature presents significant advancements in dynamic selection for regression, yet opportunities remain for more integrated, multi-criteria evaluations and standardized comparisons across methods.

Although predictive performance is the primary focus in most reviewed studies, computational cost and scalability remain underexplored dimensions. Some studies do highlight this issue: for instance, Bueno et al. (2020) observed that deep architectures such as MLPs incur high training costs, leading to less frequent model updates when compared to lightweight online learners like OS-ELMs (Online Sequential Extreme Learning Machines). Similarly, Paim and Enembreck (2023) emphasized that conventional ensemble regressors like ARF-Reg (Adaptive Random Forest for Regression) and KNN can become computationally prohibitive in streaming contexts, prompting the proposal of parallelized and more efficient alternatives like ALR (Adaptive Linear Regression). Oliveira et al. (2022) also acknowledged the cost associated

with dynamically tuning DS (Dynamic Selection) parameters per dataset to maintain high accuracy. Despite these isolated efforts, few studies conduct formal benchmarking of runtime, memory usage, or scalability across datasets of varying complexity. Future work would benefit from adopting standardized computational cost assessments to enhance the applicability of dynamic ensemble strategies in real-world systems, particularly those with time or resource constraints.

4.9. Linking research questions to outcomes

The Results and Discussion section systematically addressed all six Research Questions (RQs) formulated in this study. Specifically:

- **RQ1** was answered in Section 4.2, identifying the individual models used in dynamic selection frameworks;
- **RQ2** was addressed in Section 4.3, detailing the metrics adopted to evaluate model competency across different data regions;
- **RQ3** was explored in Section 4.4, presenting the datasets utilized and their applications;
- **RQ4** was discussed in Section 4.5, categorizing the dynamic selection strategies employed;
- **RQ5** was covered in Section 4.6, outlining the methods proposed to promote ensemble diversity;
- **RQ6** was examined in Section 4.7, describing the criteria and metrics used for performance evaluation.

This structured organization ensures that each research objective was comprehensively analyzed. It strengthens the logical connection between the research questions, the systematic review protocol, and the synthesized findings, enhancing the transparency and traceability of the review process.

5. Threats to validity and limitations of study

5.1. Critical assessment of methodological limitations in primary studies

In addition to assessing threats to the validity of this systematic review, it is important to recognize the methodological limitations observed across the selected primary studies.

Several studies lacked comprehensive experimental protocols, with limited information regarding data partitioning strategies, model hyperparameter configurations, or validation methods. This lack of detail potentially compromises the reproducibility of their findings. Additionally, although prediction accuracy was the most frequently reported evaluation criterion, few studies simultaneously considered computational efficiency, interpretability, or scalability, limiting the generalizability of their conclusions to real-world applications.

Another notable limitation concerns dataset selection and diversity. While some studies employed a wide range of datasets to validate their approaches, others relied on a very small number of datasets, often without proper justification for their representativeness. This practice raises concerns about overfitting results to specific data characteristics and challenges the robustness of the proposed techniques.

Furthermore, a subset of studies presented limited statistical validation of their results. Few applied rigorous statistical significance tests to compare dynamic selection methods against baselines, which hinders the reliability of their performance claims.

Acknowledging these methodological limitations in the primary studies provides a more nuanced perspective on the overall findings of this review and highlights important opportunities for future research to adopt more robust and transparent evaluation practices.

5.2. Threats to validity of this review

This section discusses the potential threats that could impact the results of this Systematic Literature Review, as highlighted by Petersen et al. (2015) and Wohlin et al. (2024). It outlines the strategies adopted to address and mitigate these threats while exploring the limitations of the study.

To analyze these threats, we followed the guidelines proposed by Wohlin et al. (2024), which consider four main dimensions: construct validity, external validity, internal validity, and conclusion validity.

Construct Validity ensures that the researcher uses appropriate measures for the concept under study (Petersen et al., 2015; Wohlin et al., 2024). To mitigate this threat, this study adhered to the six-step guideline proposed by Kitchenham and Charters (2007), as illustrated in Fig. 2.

The search string was carefully designed based on the research questions and used to identify relevant studies. However, we acknowledge the possibility that some studies may have been overlooked, particularly those that did not include relevant keywords in their titles, abstracts, or keyword lists. While this limitation is rare, it represents a potential threat to validity.

The study selection process, detailed in Fig. 3, was thoroughly documented, starting with an initial total of 254 studies and narrowing down to 21 primary studies. Furthermore, the authors acted as external reviewers, revising and validating the research protocol, which significantly reduced the threat to construct validity.

To enhance transparency, Table 4 details the search strings used in each database. This documentation ensures that the data collection methods align with the research questions, allowing for replication and improving the understanding of the study by other researchers.

External Validity relates to the generalization of the SLR results. According to Petersen et al. (2015); Wohlin et al. (2024), a clear description of the research process is essential to assess the degree of generalization. By following the six-step guideline proposed by Kitchenham and Charters (2007), this study mitigated threats to external validity.

Although five bibliographic sources were used to maximize the collection of relevant studies, the focus on specific databases may limit the scope of the results. Relevant studies may have been excluded if they were not indexed in the selected sources. Consequently, while the findings are applicable in certain contexts, their generalization to other domains or areas may be constrained.

Moreover, the restricted scope of databases could affect the extrapolation of results, potentially omitting studies published in alternative sources, such as specialized conference proceedings or gray literature. These limitations emphasize the importance of interpreting the findings within the context of the selected sources to avoid overgeneralization.

To enhance external validity, future reviews could adopt a more inclusive approach, incorporating additional sources and exploring methods such as manual searches or snowballing to include complementary studies.

Internal Validity refers to ensuring that the relationships observed in the study are not influenced by unmeasured factors or elements beyond the researcher's control. In this study, the methodology followed the systematic protocol proposed by Kitchenham and Charters (2007), with clear and well-defined inclusion and exclusion criteria.

Regular meetings were conducted among the authors to identify and discuss potential sources of bias. This collaborative approach helped implement strategies to minimize biases, thereby strengthening the reliability and internal consistency of the results.

Conclusion Validity addresses the accuracy of interpretations made based on the analyzed data. Although eliminating this risk entirely is not feasible, several measures were implemented to mitigate it. For example, data extraction from the primary studies was conducted by multiple authors, reducing the likelihood of individual bias and promoting balanced interpretations of the findings.

The use of visualizations, tables, and active involvement of all authors contributed to presenting the results in a clear and structured manner. These strategies enhanced the transparency of the conclusions and increased the reliability of the insights derived from the study.

By addressing threats to validity across the four dimensions–construct, external, internal, and conclusion–this study demonstrates a commitment to methodological rigor and transparency. While limitations such as the restricted scope of databases and potential omissions were acknowledged, the strategies implemented provide a solid foundation for the reliability and generalizability of the results. Future work should consider extending the scope of data sources to further strengthen the validity of similar studies.

6. Conclusion

This Systematic Literature Review consolidates the state of the art on dynamic selection techniques applied to regression tasks using ensemble models. The analysis of 21 primary studies published between 2015 and 2024 reveals a research field in active expansion, driven by the need to improve adaptability and accuracy in nonstationary and heterogeneous environments.

The findings demonstrate that dynamic strategies–such as Dynamic Selection (DS), Dynamic Weighting (DW), and hybrid approaches like META-DES–consistently outperform static ensembles by tailoring model behavior to localized patterns in the data. These approaches have proven especially effective in domains such as healthcare, energy forecasting, finance, and software engineering, where evolving data distributions demand models capable of continuous adaptation.

This review offers several contributions to the field. First, it maps a diverse landscape of 59 regression models and 26 evaluation metrics used across 144 datasets, highlighting methodological breadth and domain applicability. Second, it proposes a structured taxonomy of dynamic selection strategies, clarifying conceptual boundaries between DS, DW, DWS, and optimization-based approaches. Third, it identifies key implementation practices, including the use of heterogeneous ensembles, adaptive sliding windows, residual modeling, and meta-learning layers to enhance prediction performance.

Despite these advances, critical challenges remain. Computational cost, limited scalability to real-time or high-dimensional settings, and the lack of standardized evaluation protocols hinder broader adoption in practical scenarios. In this context, future research should aim to: (i) design computationally efficient frameworks suitable for data streams and large-scale applications; (ii) integrate dynamic selection with deep learning and decentralized architectures; (iii) incorporate interpretability and transparency into model selection processes; and (iv) establish benchmarks and evaluation standards to enable fair and reproducible comparisons.

By advancing in these directions, dynamic selection techniques can evolve into scalable, reliable, and interpretable solutions capable of supporting complex regression problems in real-world environments.

CRediT authorship contribution statement

Halcyon Davys P. de Carvalho: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft, Writing - review & editing; **João Fausto L. de Oliveira:** Formal analysis, Writing - review & editing; **Roberta Andrade de A. Fagundes:** Supervision, Formal analysis, Writing - review & editing.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. References for the 21 included SLR papers [PS01-PS21]

[PS01] Fuzzy Inference Systems Applied to the Combining Regression Models

[PS02] On the Evaluation of Competence Measures for Time Series Forecasting

[PS03] Dynamic ensemble mechanisms to improve particulate matter forecasting

[PS04] Dynamic Regressor/Ensemble Selection for a Multi-Frequency and Multi-Environment Path Loss Prediction

[PS05] Interpretable Dynamic Ensemble Selection Approach for the Prediction of Road Traffic Injury Severity: A Case Study of Pakistan's National Highway N-5

[PS06] Meta-learning recommendation of default size of classifier pool for META-DES

[PS07] Heterogeneous Ensemble Dynamic Selection for Software Development Effort Estimation

[PS08] A weighted distance-based dynamic ensemble regression framework for gastric cancer survival time prediction

[PS09] A Hybrid System Based on Dynamic Selection for Time Series Forecasting

[PS10] Boosting the prediction of molten steel temperature in ladle furnace with a dynamic outlier ensemble

[PS11] Forecasting of Solar Energy Generation via Dynamic Model Ensemble

[PS12] A SVR-based ensemble approach for drifting data streams with recurring patterns

[PS13] Ensemble Effort Estimation using dynamic selection

[PS14] MINE: A framework for dynamic regressor selection

[PS15] Improving the accuracy of long-term travel time prediction using heterogeneous ensembles

[PS16] Evaluating Competence Measures for Dynamic Regressor Selection

[PS17] Solar Irradiance Forecasting Using Dynamic Ensemble Selection

[PS18] Advances on Concept Drift Detection in Regression Tasks Using Social Networks Theory

[PS19] Adaptive Linear Regression for Data Stream

[PS20] A dynamic ensemble method for residential short-term load forecasting

[PS21] Dynamic ensemble of regression neural networks based on predictive uncertainty

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